Léon MORALES

Generative AI Engineer — LLM & Graph RAG

Full-time or VIE • Available from April 2027 • Open to relocation worldwide

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PROFILE

Dual-degree AI engineer (UTBM / UQAC) building LLM-based systems end to end: Graph RAG on Neo4j (knowledge graph construction, graph propagation), local LLMs and LLM-as-a-judge evaluation, through to cloud deployment (FastAPI, Docker, Cloud Run, Terraform, CI/CD). R&D experience in drug discovery and academic research. Neo4j Certified Professional.

EXPERIENCE

Generative AI Engineer (Graph RAG) — End-of-studies internship Sept. 2026 – Feb. 2027
Astek Group, Innovation Lab (R&D) • *confidential project* La Défense, France

- Designing a **Graph RAG** platform in **Python** for documents too long for an LLM's context window, with automatic construction of a **Neo4j** knowledge graph.
- **Graph propagation** method to select the most relevant context for the LLM.
- Local, private LLMs (**Ollama**) so that no data leaves the company; **LLM-as-a-judge** evaluation.

Research project — Human Activity Recognition Jan. – May 2026
LIARA Lab (Ambient Intelligence), UQAC Chicoutimi, Canada

- Designed a **self-supervised contrastive learning** architecture for wearable sensors (IMU/PPG), aiming to outperform state-of-the-art involution networks.
- Literature review, **PyTorch** implementation, experiments with about twenty participants; paper under revision.

AI Engineering Intern Sept. 2024 – Feb. 2025
KSILINK (biotech, drug discovery) Strasbourg, France

- **Graph RAG chatbot:** conversational agent linking an open-source LLM to a **Neo4j** database (100k+ nodes), deployed for about fifteen biologists.
- **R&D:** implemented a **multimodal contrastive learning** model (cell images and molecular structures) based on *Cell Morphology-Guided Small Molecule Generation with GFlowNets*.
- **Computer Vision:** CNN (**PyTorch Lightning**) classifying 4 cell types on noisy data, ≈ 85 % accuracy.

President of the SKI'UTBM Association July 2023 – Feb. 2024
UTBM Belfort, France

- Ski trip for 200 students: managed a €100k+ budget and a team of 11 people.

PROJECTS

Graph RAG Assistant in production — *my portfolio's chatbot, since 2025* • *website* • *code*
Text-to-Cypher API (read-only) on Neo4j with Gemini and automatic model fallback; Cloud Run, Terraform infrastructure, keyless GitHub Actions CI/CD, rate limiting and cost alerts. *Python, FastAPI, Neo4j, Docker, GCP, Terraform.*

RAG Assistant & Dashboard — *Conformit X UQAC Hackathon, 2nd place (25 h, team of 4), Nov. 2025* • *code*
Industrial risk management assistant: RAG pipeline translating natural language into secure SQL. *Python, Gemini, SQL, FastAPI, Streamlit, Docker.*

EDUCATION

Master's Degree in Computer Science (AI profile) — Dual degree 2025 – 2027 (expected)
Université du Québec à Chicoutimi (UQAC) Chicoutimi, Canada

Master of Engineering in Computer Science — Data Science & AI 2021 – 2027 (expected)
Université de Technologie de Belfort-Montbéliard (UTBM) Belfort, France

Exchange semester — Computer Engineering (in English), *Yeditepe University*, Istanbul Jan. – June 2023

SKILLS

- **Generative AI & LLMs:** Ollama, Hugging Face Transformers, Gemini API; RAG / Graph RAG, Text-to-Cypher / SQL, LangChain, LLM-as-a-judge, embeddings, ChromaDB, fine-tuning (LoRA)
- **Machine Learning:** PyTorch (Lightning), TensorFlow, Scikit-learn, XGBoost, Pandas; contrastive learning, NLP, Computer Vision
- **MLOps & Cloud:** FastAPI, Docker, Google Cloud (Cloud Run), Terraform, GitHub Actions (CI/CD), pytest, Weights & Biases
- **Languages & tools:** Python (advanced), SQL (PostgreSQL), Cypher, C/C++, Java; Git, Neo4j, Streamlit
- **Certifications:** Neo4j Certified Professional • NVIDIA DLI — Accelerating End-to-End Data Science Workflows
- **Spoken languages:** French (native), English (C1, professional), Spanish (B1)
- **Interests:** travel (15+ countries), guitar (7 years), skiing, snowboarding, basketball